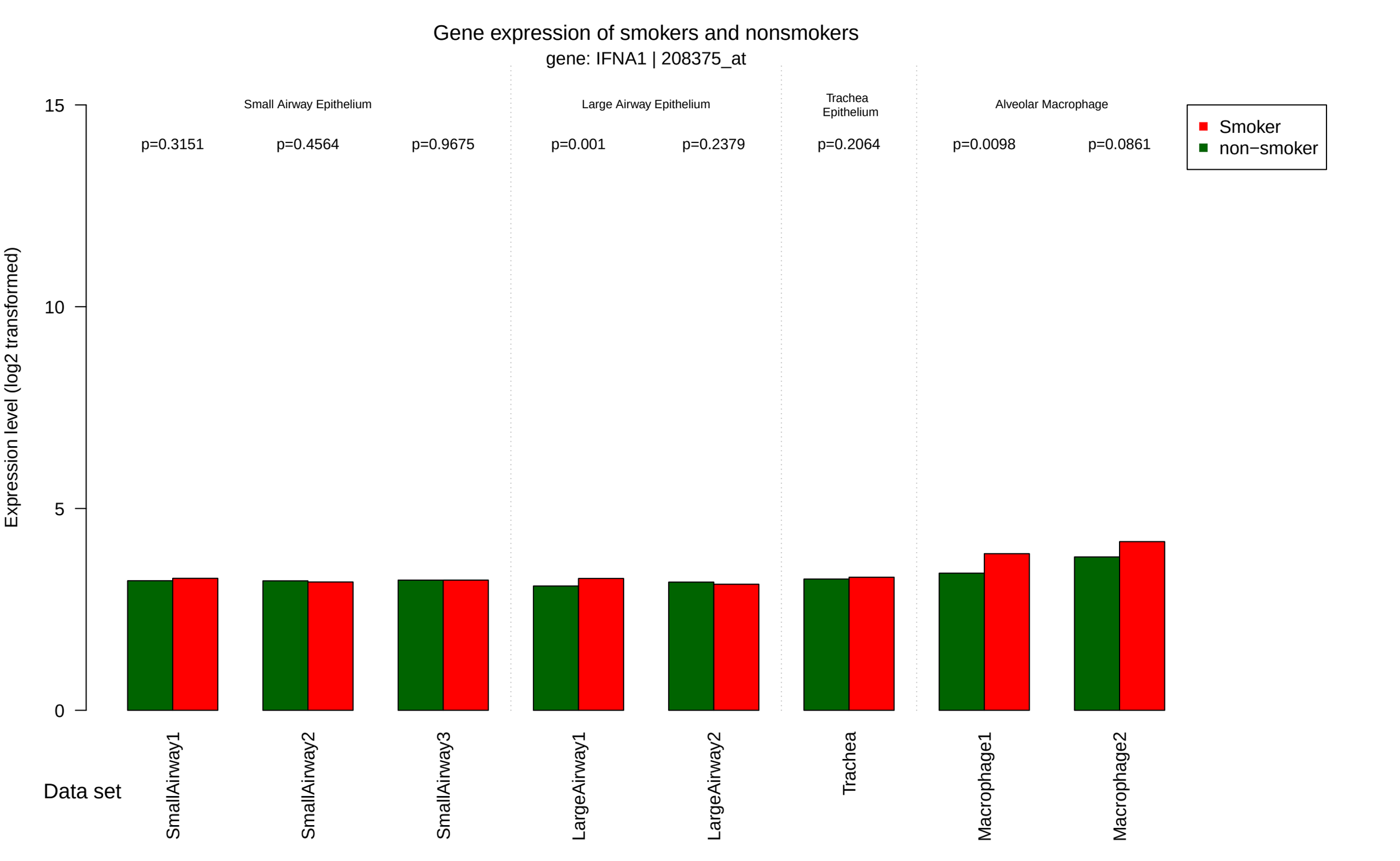
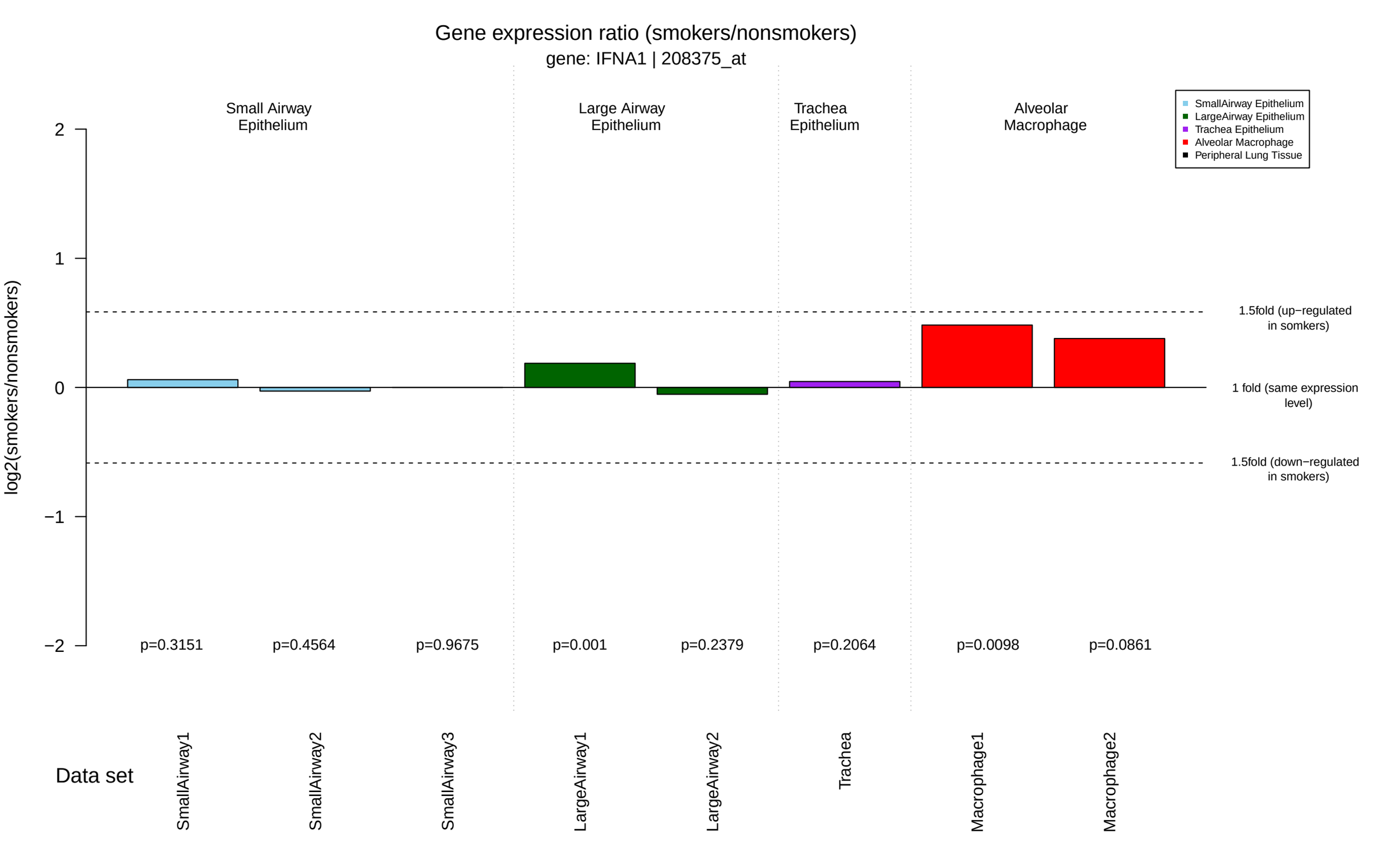
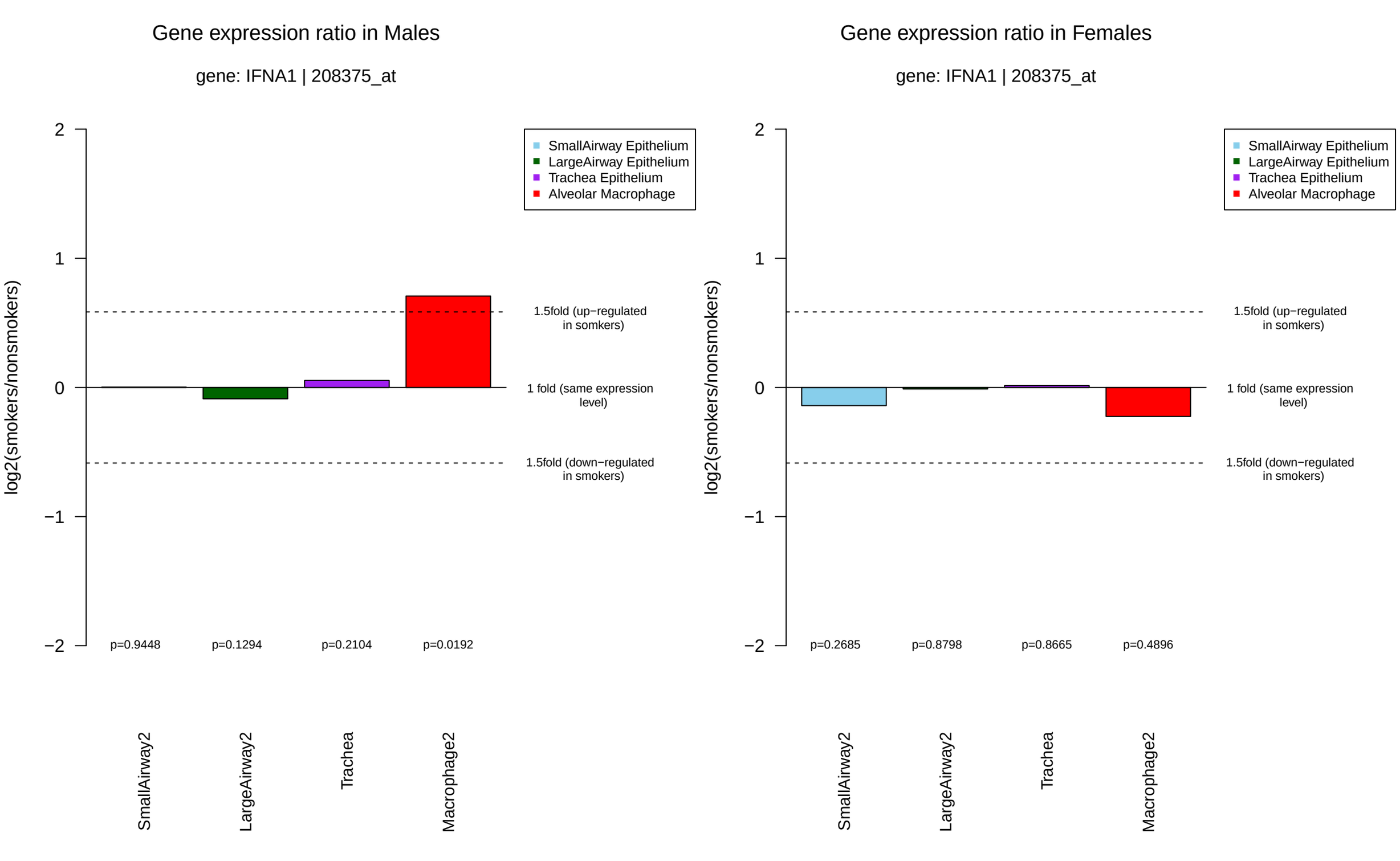
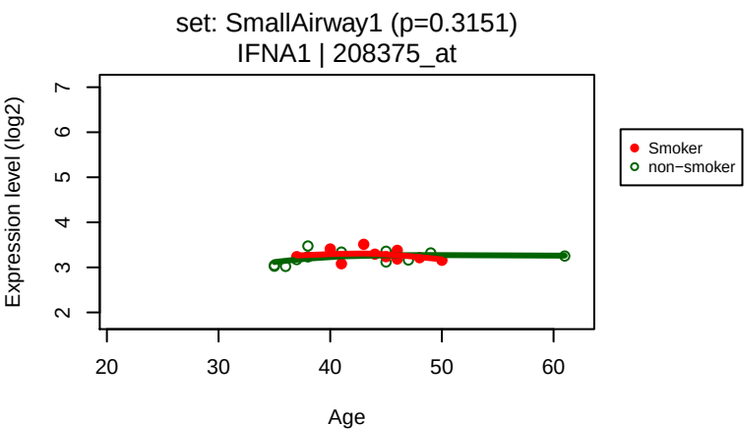
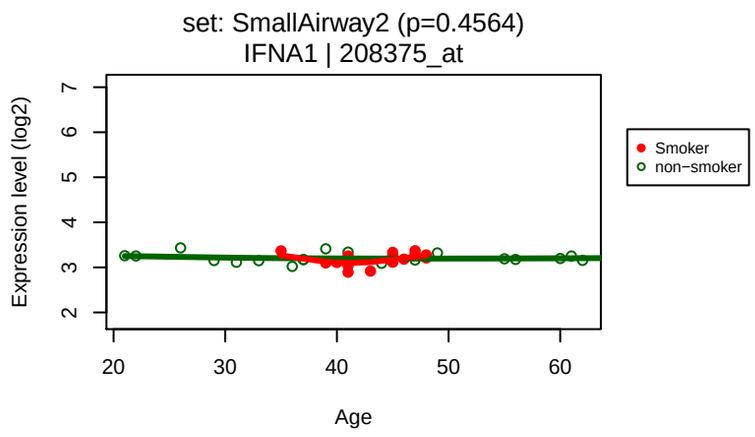
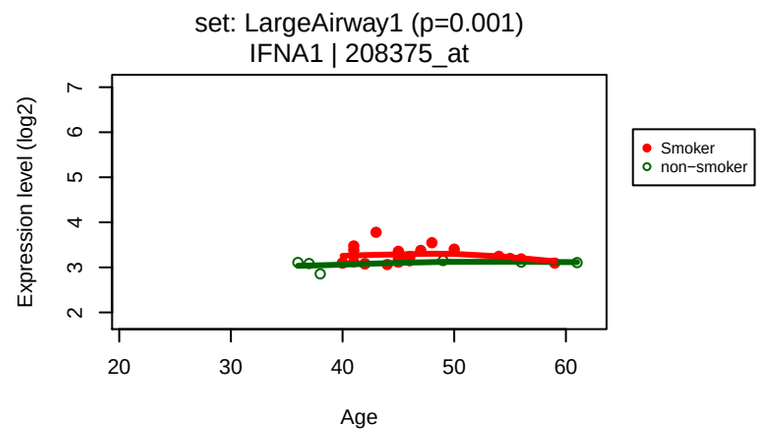

Gene expression of smokers and nonsmokers
gene: IFNA1 | 208375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3151
p=0.4564
p=0.9675
p=0.001
p=0.2379
p=0.2064
p=0.0098
p=0.0861
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

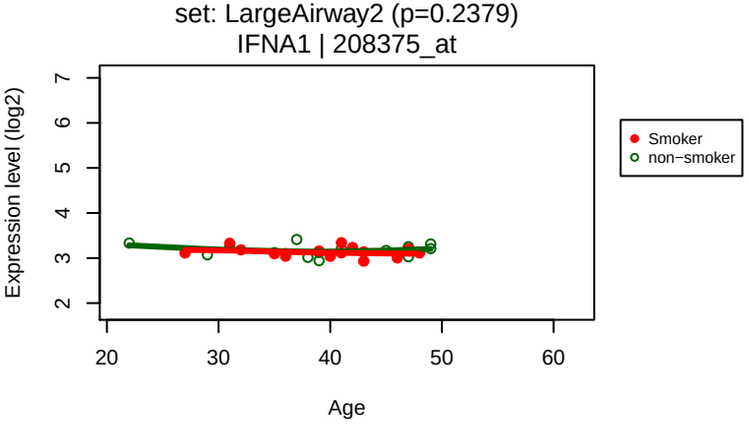

Gene expression ratio (smokers/nonsmokers)
gene: IFNA1 | 208375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3151
p=0.4564
p=0.9675
p=0.001
p=0.2379
p=0.2064
p=0.0098
p=0.0861
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

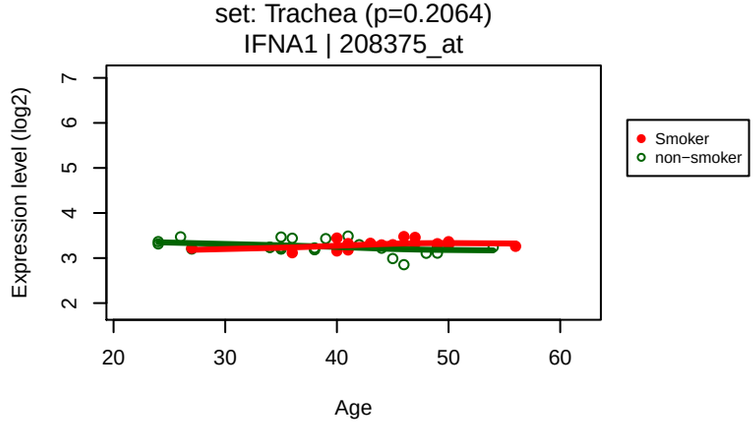

Gene expression ratio in Males
gene: IFNA1 | 208375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9448
p=0.1294
p=0.2104
p=0.0192
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IFNA1 | 208375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2685
p=0.8798
p=0.8665
p=0.4896
SmallAirway2
LargeAirway2
Trachea
Macrophage2

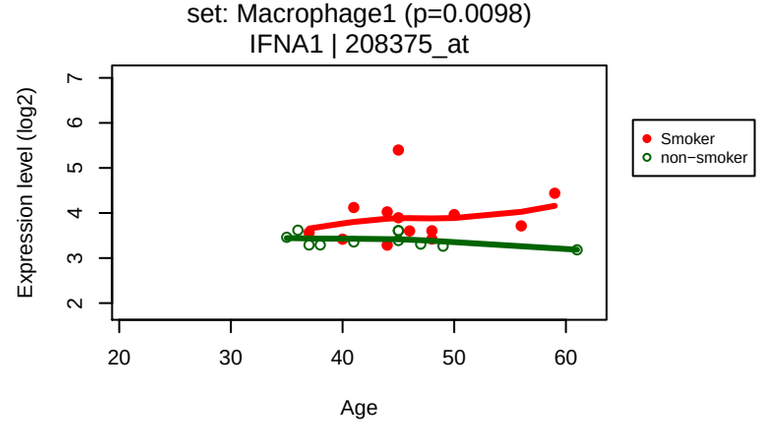

set: SmallAirway1 (p=0.3151)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker

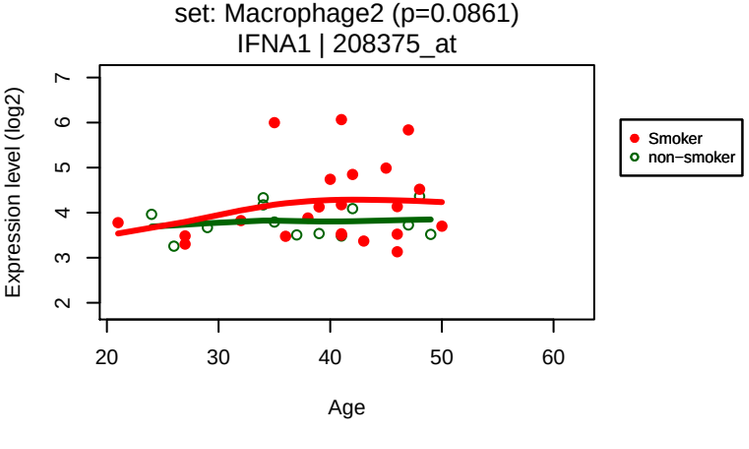

set: SmallAirway2 (p=0.4564)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker

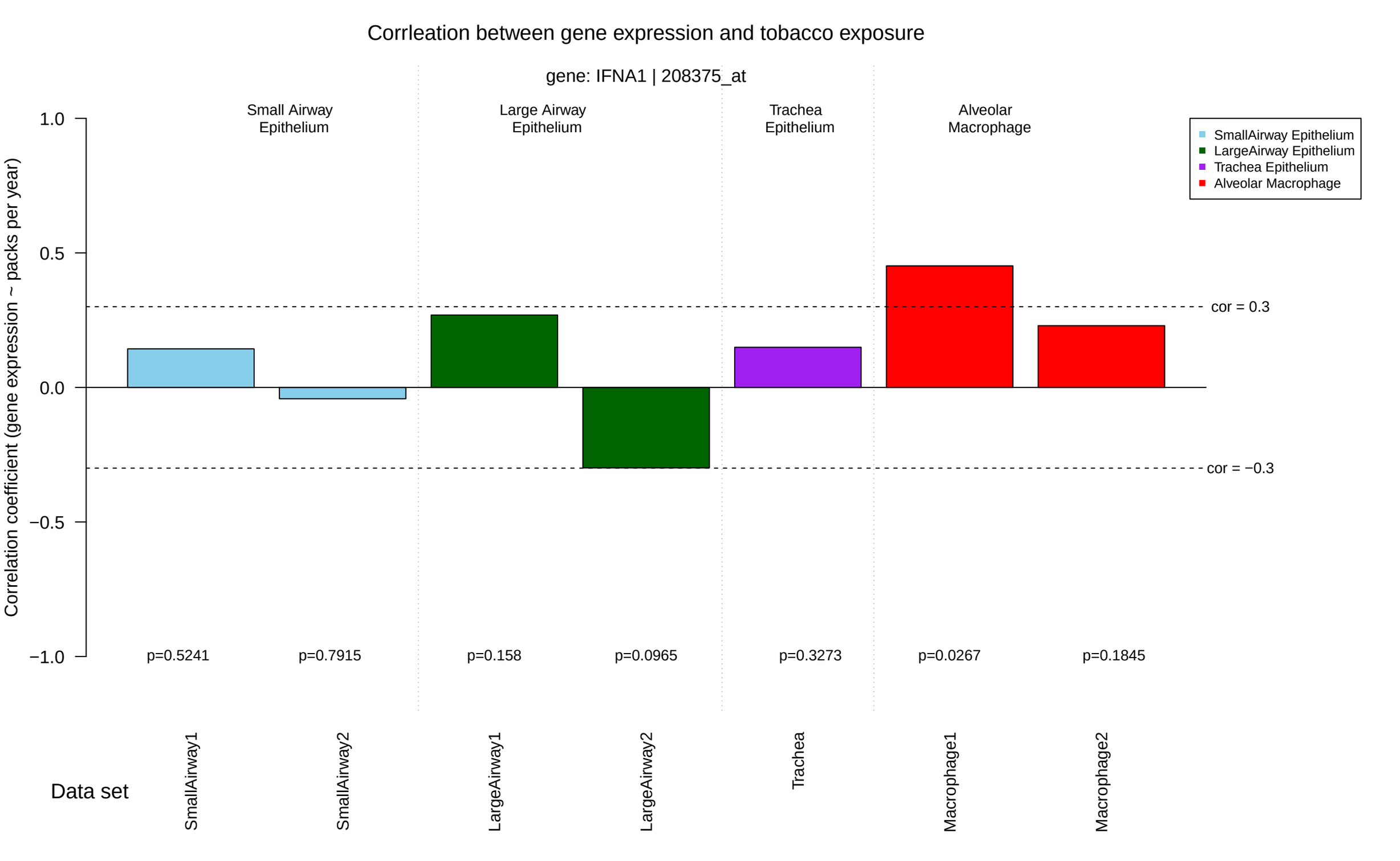

set: LargeAirway1 (p=0.001)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2379)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2064)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0098)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0861)
IFNA1 | 208375_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IFNA1 | 208375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5241
p=0.7915
p=0.158
p=0.0965
p=0.3273
p=0.0267
p=0.1845
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2